macOS Native Circuit Simulator (MultiSim) - Architecture & Implementation Plan

1. Overview

MultiSim for Mac is a native macOS schematic capture and circuit simulation application built with **Swift** and **SwiftUI**. It combines:

1. **Interactive Real-Time Mode:** 60/120 FPS live simulation with animated electron/current flow, live switch toggling, and real-time virtual oscilloscope traces (similar to Falstad / Multisim Live).
2. **Scientific SPICE Engine:** Exact Modified Nodal Analysis (MNA), companion models for reactive components (C, LC, L), Newton-Raphson non-linear iterations for semiconductors (D, QD, Q), and standard SPICE netlist (`.cir`) generation/export.

2. System Architecture

3. Step-by-Step Implementation Roadmap

Step 1: Core Mathematical Simulation Engine (MultiSimEngine)

- ✓
  Implement matrix data structures and high-performance linear equation solver ($A \cdot x = b$ A · x = b) using LU decomposition with partial pivoting.
- ✓
  Implement Modified Nodal Analysis (MNA) stamp formulations:
  - Conductors / Resistors ($G = 1/R$G = 1/R)
  - Independent DC and AC Voltage Sources (adds auxiliary current variables)
  - Independent Current Sources
  - Ground reference (Node 0)

- ✓
  Implement Reactive Companion Models (Numerical Integration):
  - Capacitors: $G_{eq} = C/\Delta t$Geq = C/Δt, $I_{eq} = (C/\Delta t) \cdot v_{prev}$ Ieq = (C/Δt) · vprev
  - Inductors: $G_{eq} = \Delta t/L$Geq = Δt/L, $I_{eq} = i_{prev}$ Ieq = iprev
- ✓
  Implement Newton-Raphson iteration for non-linear components (Diode Shockley model, LEDs).
- ✓
  Build automated unit test suite verifying:
  - Ohm's law & voltage divider (Passed)
  - RC transient charging curve $\tau = R \cdot C$τ = R · C (Passed)
  - Diode rectification curve (Passed)
  - SPICE netlist export (Passed)

Step 2: Circuit Graph & Document Data Model

- ✓
  Define `CircuitComponent`, `Pin`, `Wire`, and `Circuit` data structures.
- ✓
  Automatic Netlist extraction: Union-Find algorithm mapping pins to electrical node indices.
- ✓
  SPICE Netlist Exporter (`.cir` format): standard syntax output compatible with ngspice / LTspice.
- ✓
  Preset circuits (AC Rectifier, RC Filter, LED Switch).

Step 3: Interactive Schematic Canvas (SwiftUI)

- ✓
  Infinite 2D dot grid with smooth pan & pinch-to-zoom.
- ✓
  Component placement, dragging, 90° rotation (`R` key), deletion (`Backspace`).
- ✓
  Smart Manhattan (orthogonal) wire routing between component pins.
- ✓
  Auto-junction solder dots at wire T-intersections and crossings.
- ✓
  Native macOS electronic CAD styling (crisp schematic symbols).

Step 4: Live Real-Time Simulation & Visual Effects

- ✓
  Simulation ticker running at 60 FPS.
- ✓
  Animated moving current dots along wires (speed & direction tied to calculated current).
- ✓
  Real-time LED forward current glow effect.
- ✓
  Interactive components: clickable switches (toggle open/closed live).

Step 5: Virtual Instruments

- ✓
  Dual-Channel Oscilloscope:
  - Real-time scrolling waveform display.
  - Channel A (Red) and Channel B (Cyan) probe assignment.
  - Time/Div and Volts/Div controls.
- ✓
  Digital Multimeter / Inspector:
  - Floating probe to read DC voltage, AC RMS, and current across any component.

Step 6: Extended Component Library

- ✓
  Active devices: BJT Transistors (NPN / PNP).
- ✓
  Integrated circuits: Ideal & LM741 Op-Amp model.
- ✓
  Digital logic gates: AND, OR, NOT, NAND, NOR, XOR.